MODULE 3

HOW DO I VALUE A BOND?

BOONE BOWLES

Agenda

1. What do we remember from online bootcamp?
2. How do I value a bond?
3. How are interest rates determined?
4. What do I need to remember from all this?



Can we remember how to do the harder questions?

1. What is $85 million today worth 16 years from now? (The interest rate is 6 %).

2. What interest rate would turn $100 million into $260 million in 7 years?

3. What interest rate would turn $100 million into $260 million in 17 years?

4. What is the future value 8 years from now of $100 million to be received next year, $200 million the year after that, and $300 million two years after that? (The interest rate is 5.25 %).

5. What is the present value of that same stream of cash flows?

Can we remember how to do the harder questions?

6. Let's imagine this interesting cash-flow stream:
   - Right now, there will be a cash outflow of $700 million.
   - Next year, there will be another outflow of $700 million.
   - The year after that will have no cash inflow or outflow.
   - The year after that will see a cash inflow of $100 million.
   - The next two years will be the same.
   - Then, the annual cash flow will be $120 million for three straight years.
   - Then, the annual cash flow will be $160 million for three straight years.
   - After this, the annual cash flow is expected to grow a bit faster than the U.S. economy (≈ 3 %).
   - Discount rate = 9 %.

What is the Present Value?

How do we characterize bonds?

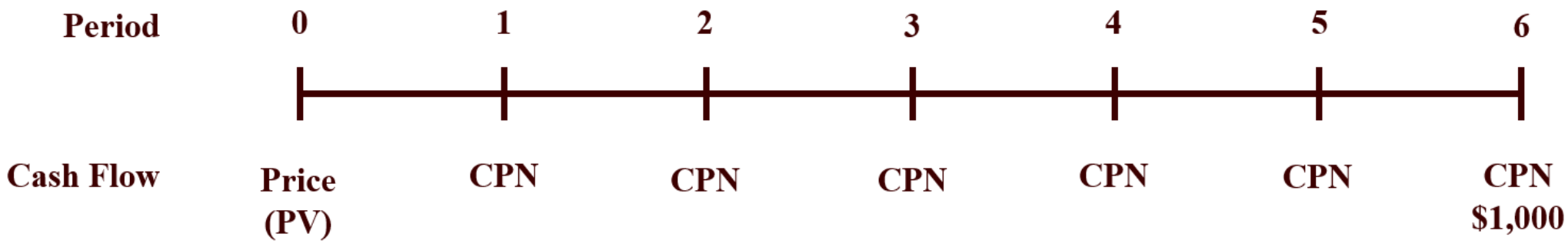
Bonds are simply streams of cash flows… we can value those no problem.

Term or Maturity	→	Time to final principal payment
Coupons	→	Periodic interest payments
Face Value	→	Payment at maturity ($1000)
Price	→	PV of future cash flows
Yield to Maturity (ytm)	→	Interest rate equating price with cash flows

REMEMBER: If you're struggling understanding bonds

- They are loans that a company must pay back to the bank
- Except in this case the bank is the market.

How do we characterize bonds?





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